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Temperament and the Development of Personality

Mary K. Rothbart and Stephan A. Ahadi

We present a psychobiological approach to personality development, incorporating developmental principles outlined by R. B. Cairns (1979). We review individual differences in temperament and ask how a temperamental approach to personality might be congruent with these complex and flexible principles. We then illustrate the nature of temperamental processes by considering several developmental topics. We first consider the interaction of infant distress-proneness and maternal behavior in the development of attachment. We then describe the development of self-regulatory mechanisms, emphasizing development of conscience, aggression, and mastery motivation. Finally, we briefly review mechanisms of temperament and environment interaction, illustrating these processes through variable developmental pathways for risk of adolescent and adult psychopathology. Throughout, we stress the idea that temperamental models of personality development are dynamic, interactive, and fit well with Cairns's developmental principles.

Psychology textbooks frequently describe three approaches to personality and social development: psychoanalytic, social learning, and cognitive developmental. These theories are often described as opposing or conflicting with one another in accounting for the development of individual differences. Theoretical and research approaches to personality and social development are now being implemented, however, that neither depend on a conflict approach nor require the researcher to "take theoretical sides" in order to study development. These changes in the field were foreshadowed in a now classic textbook on social development written by Robert Cairns in 1979. In his book, Cairns put forward what he called the *developmental synthesis.* This synthesis does not place theories in opposition to each other but describes their complementary contributions in relation to a psychobiological analysis.

In this article, we review some of the principles of a psychobiological approach, applying them to the study of personality development. In doing so, we employ a temperamental perspective on personality development that from ancient times has been psychobiological. The major purpose of this article is to encourage the reader to think about how a complete understanding of personality and psychopathology must be a developmental one, as Freud and others have argued, and to demonstrate with research examples some of the ways in which issues in personality can be approached by following a developmental analysis. It is not possible in this limited space to provide a comprehensive review of temperament and child development (for such a review see Rothbart, 1989b). Rather, our goal is to demonstrate, with research examples, how a temperament model of development conforms to the dynamics required of the developmental synthesis.

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The Psychobiological Approach and Temperament

We begin by considering six guiding principles of the psychobiological approach to development as presented by Cairns (1979, pp. 325–326):

> 1. Behavior, whether social or nonsocial, is appropriately viewed in terms of an organized system . . .
> 2. The system of which behavior is a part is not merely "organismic" but for some functions, particularly social behaviors, it must be expanded to include the acts of other organisms and the reciprocal relationships that are formed with them . . .
> 3. There is a continuity in development, such that the organization at one stage provides the basis for organization at the next succeeding stage . . . development is essentially a dynamic process that promotes reorganization and adaptation across time . . .
> 4. The need for multiple levels of analysis . . . is a direct corollary of the assumption that there are interlocked systems associated with the control of behavior, from neurobiological events to sociocultural ones.
> 5. Behavior should be viewed in terms of the organization of which it is a part . . . What appears to be the "same" activity at two different stages or in two different species may be similar in only superficial properties . . .
> 6. The organism is continuously active and adaptive throughout the course of development . . . (with adaptation) relative to what system is involved and the circumstances in which adaptation is assessed.

Although not explicitly developmental, many of these principles were central to conceptions of personality as set forth by Allport, Murray, and other prominent founders of the field of personality (for a brief review see Pervin, 1990). In this article, we apply these principles to issues of personality development, adding social learning, cognitive developmental, and psychoanalytic contributions to the synthesis as we continue our exposition. We begin with a brief discussion of individual differences in temperament and ask how a trait-based approach to personality, such as temperament, might possibly be congruent with the complex and flexible view of development expressed above.

We define temperament as constitutionally based individual differences in reactivity and self-regulation, influenced over time by heredity, maturation, and experience. By reactivity we

mean responsiveness of emotional, activation, and arousal systems, and by self-regulation we mean processes such as approach, avoidance, and attention that serve to modulate the reactivity of the individual (Rothbart & Derryberry, 1981; Rothbart & Posner, 1985). We see the young infant as highly reactive. With increasing development of both emotional (fear and shyness) and attentional controls, interaction among temperamental characteristics becomes increasingly likely (Rothbart, 1989b).

How do Cairns's principles apply to temperament constructs? First, we know that since the time of the early Greco-Roman theorists, temperament has involved multiple levels (Principle 4). The ancient fourfold typology was described in terms of human physiology as it was understood at that time in the bodily balance of the humors: the sad and anxious melancholic's overabundance of black bile, the irritable choleric's preponderance of yellow bile, the extraverted sanguine's blood, and the unreactive phlegmatic's phlegm. More recently, temperamental dimensions have been related to a nervous system that is increasingly recognizable to us. Eysenck (1967), for example, related extraversion to arousal systems, and neuroticism to limbic reactivity. Currently, dimensions of positive affect and approach (extraversion) have been related to dopamine systems, fearfulness to the workings of the amygdala and the septohippocampal system, and attention to posterior and anterior brain networks (see Depue & Iacono, 1989; Gray, 1982; LeDoux, 1987; Posner & Peterson, 1990; and for reviews, Gunnar, 1990; Rothbart, 1989a; and Rothbart, Derryberry, & Posner, in press).

As in the area of cognitive neuroscience, an affective neuroscience approach linking temperament and neurophysiology allows different levels of analysis to inform each other. Evidence from neuroscience makes it clear to us that the individual differences we study are part of the whole organism's functioning (Principle 1). At the same time, temperament functions within a social context, influencing and being influenced by other people (Principle 2). The study of temperament is located at a level of analysis with physiology on one side and social interaction on the other, allowing a particularly good vantage point for thinking about how personality develops.

Rather than considering temperament as an unchanging set of entities within the person, it is probably better for psychologists to simply think about temperament as a set of variables, measured by aggregating individual response across multiple situations. A fearful infant may react with distress or behavioral inhibition (or both) to a new adult, child, dog or cat, to the vacuum cleaner, the blender, or a stairway, and we aggregate shared variance across different situations to get at a temperamental "trait." Note that in referring to multiple situations we do not mean all situations; a trait refers to shared variance across situations with the needed eliciting conditions for the reaction. Measuring fearfulness at a given age for a given child also makes no statement that it will remain unchanged from one age to another, even in rank ordering on the measures. Depending on the period of development studied, we may expect stability or change in our measure from one age to another. Once measured, however, individual differences on this or other temperament variables allow us to pose important questions. We can ask how temperamental organization at a particular time in development provides a basis for reactions to and adaptations toward environmental events (Principles 3 and 6). We can also ask how these reactions and adaptations take place within a social context (Principle 2). This is not to imply a general instability of temperament across time. Assessments of temperament-related variables show remarkable stability across time (e.g., Block, 1993) and especially so in the adult years (McCrae & Costa, 1990). Our point, rather, is that temperamental characteristics are not static and unchanging but adapt to environmental demands.

Temperament in Infancy

If we view temperament as a set of constitutionally based individual-differences variables, we can then look into the research literature to find out what has been learned to date about early temperamental variability. This analysis is important to an understanding of personality development in several ways. First, we expect early temperament to provide the within-the-person substrate from which personality develops. The personality domain contains much more than temperament, however, including patterns of habitual behavior, skills, and the content of individual thought, values, needs, and goals. It also includes the perception of self, others, and events. Temperament has a role in influencing the development of these more content-related variables, as we suggest below. Reciprocally, the way we think about ourselves and others arouses our emotions and often leads to additional adaptations. These adaptations develop out of the interaction between temperament and experience and may involve coping styles or defenses.

Third, even in beginning to consider the temperament dimensions now emerging from infancy research, we can see similarities with the higher order factor structures that have been repeatedly extracted from adult studies of personality. For example, in the newborn, individual differences in distress-proneness can be observed in neonatal behavioral assessments, and these individual differences demonstrate some stability across the first two years of life (Rothbart, 1989a). Other dimensions observed in the newborn period include activity level, attentional orienting, and alertness. Although approach and withdrawal behaviors are observable in the newborn, to date they have not been assessed as temperament variables.

Studies of temperament in infants 2–3 months of age and older have their origins in Thomas and Chess's (1977; Thomas, Chess, Birch, Hertzig, & Korn, 1963) nine dimensions of temperament or in other psychobiologically based sets of variables (Buss & Plomin, 1975, 1984; Rothbart, 1981). From these studies, five to six dimensions have emerged from item-level factor-analytic and other research (see review by Rothbart & Mauro, 1990). These dimensions include two kinds of negative reactivity: (a) fearfulness and (b) irritability or anger and frustration; one positive affect variable: (c) approach or positive affect; along with (d) activity level and (e) attentional persistence. A small (f) rhythmicity factor also has been extracted but is not included in our discussion here.

In scale-level factor analyses of the Children's Behavior Questionnaire (Rothbart, Ahadi, Hershey, & Fisher, 1993) for older children between 3 and 8 years old, we have consistently recovered three broad factors (Ahadi & Rothbart, in press). One fac-

tor, labeled *Surgency/Extraversion,* was defined primarily by the scales of Approach, High Intensity Pleasure, Activity Level, Impulsivity, and, loading negatively, Shyness. A second factor, defined by scales of Discomfort, Fear, Anger/Frustration, Sadness and, loading negatively, Soothability, corresponds to a broad factor we labeled *Negative Affectivity.* A third dimension, defined by loadings for the scales of Inhibitory Control, Attentional Focusing, Low Intensity Pleasure, and Perceptual Sensitivity, was labeled *Effortful Control.* This basic structure has been replicated in the United States (e.g., Kochanska, DeVet, Goldman, Murray, & Putnam, 1993) and has also emerged for a sample of Chinese 6- and 7-year-olds (Ahadi, Rothbart, & Ye, in press).

The reader will note similarities between these dimensions and the "Big Five" or "Big Three" higher order factors emerging from studies of personality (Eysenck & Eysenck, 1985; Goldberg, 1993; McCrae & Costa, 1987; Tellegen, 1985; Watson, Clark, & Harkness, 1994 [this issue]). The first two dimensions of infant temperament, fearfulness and irritability, appear to correspond to the childhood and adult dimension of negative affectivity or neuroticism, that is, a general tendency to experience and express negative emotions. The infant dimension of approach or positive affect and possibly activity level may be related to the child and adult dimension of extraversion or positive affectivity, characterized by positive affect and an approach, stimulus-seeking, or reward orientation (Gray, 1982; Zuckerman, 1991). The later-developing infant dimension of attentional persistence may overlap with child and adult variation on the dimension of control or constraint, reflecting attentional self-regulatory systems that can assert control over other temperament systems through effort or will (Rothbart & Posner, 1985).

Remembering that similarity of behavior need not imply similarity of underlying process (Principle 5), we are wary of viewing these similarities as providing any complete account of personality (Ahadi & Rothbart, in press). Nevertheless, the similarities are exciting, and longitudinal work on these infant temperament variables indicates some predictability from laboratory measures in the newborn or infancy periods to measures taken in early (see review by Rothbart, 1989b) and middle childhood (Rothbart, Derryberry, & Hershey, 1993).

Temperament and Development

In the next sections of this article, we discuss more specific applications of temperament variables in the study of personality across the lifespan. Before proceeding, however, we review some theoretical ideas about the role of temperament in personality development, and as a corollary, in the development of psychopathology. This discussion is adapted in part from two of our previous articles (Rothbart, Ahadi, & Hershey, 1994; Rothbart, Posner, & Hershey, in press).

First, temperament is implicated in learning processes (Eysenck, 1967; Gray, 1982). Escalona (1968) argued that the effective experience of a given situation on the child is not the same for all, in ways that are related to temperament, and Wachs and Gandour (1983) discussed organismic specificity in children's reactions to situations. Stranger fear may be minimal for one child; another child may experience painful distress. These emotional reactions can then be directly conditioned to the context in which they occurred; they may also be related to operant responses allowing the child to avoid the problem by avoiding strangers in the future. For the older child or adult, avoidance of parties or new situations may reflect such an adaptation. These reactions will also influence how children see themselves and others. Self-perception of emotional reactivity will influence the concepts children and adults develop about their own abilities and likelihood of successful performance, their possibilities for action within a given situation, as well as the concepts they have about whether objects or events are threatening or safe, bad or good.

Second, regulation of one's own reactivity may lead to patterns of adaptation to situations that may prove at least as important to our assessment of temperament as reactivity itself. Bell (1974) made a very helpful analysis of the child's and the parent's upper and lower bounds for stimulation. An individual with a low threshold for distress will attempt to keep things quiet. An individual requiring high levels of stimulation for pleasure will attempt to increase excitement. Mismatches between individuals (e.g., parents and their children, marriage partners) in situational requirements for comfort may lead to problems in communication and cooperation. In addition, situational requirements for the experience of lower or higher levels of stimulation may lead to distress for the person whose own characteristics provide a poor match with the social and physical environment (Strelau, 1983).

Third, temperament itself follows a course of development (Plomin, DeFries, & Fulker, 1988; Rothbart, 1989b; Rothbart & Derryberry, 1981), with important implications for personality development. The newborn infant demonstrates general distress, but frustration responses are not observed until 2 months of age, and fear reactions are observed several months later. The positive, approaching 6-month-old is very different from the newborn. The hesitating, self-soothing 10-month-old is beginning to demonstrate both fear and caution (Rothbart, Ziaie, & O'Boyle, 1992). In a strange place or with novel stimuli, the 8–10-month-old shows the beginnings of the inhibition of approach and emotional expressiveness that will constitute one of two powerful temperamental control mechanisms. Development of fear control is followed by effortful inhibitory control emerging late in the first year and developing across the preschool years and beyond. Effortful control allows further modulation of approach and expressiveness according to situational demands or explicit instructions from adults. We have argued that this latter kind of control is attentional and that its development is likely connected with the development of the anterior executive attention system (Posner & Rothbart, 1992). We elaborate some of these points in the following paragraphs.

The development of temperament occurs in the context of important cognitive changes in the developing child. The development of a sense of self over the first 2 years of life leads to the possibility of shame or embarrassment over one's actions (Kagan, 1989). Later developments of the tendency to attribute stable qualities to objects, persons, and the self provide further possibilities for children to make judgments about themselves and others.

In the social area, the development of attachment means that other people and their evaluations become important to the

child, so that the child will attempt to be a valued person. This orientation may be associated with the need in the well socialized child or adult to deny responsibility for acts that are seen as not valued (Snyder, Higgins, & Stucky, 1983). In a less socialized child, there may be related problems with attachment, with the behavior of significant models, with the child's reinforcement history, with the child's self-regulatory capacities, or with some combination of these factors. Woven across development will be a pattern of reactions, internal concepts, and adaptations to significant situations and persons that can be seen to constitute the personality. The pattern will be multidetermined, but in some cases one or more of the influences alone or in interaction will be so extreme as to dominate and to make adaptations to events difficult. This situation may be related to actions, thought, and/or behaviors that are seen as psychopathological. In the following sections, we give examples of several age periods in which temperament and its development have been related to the development of personality and in some cases to the risk for psychopathology.

Distress-Proneness and Attachment

Research on human attachment continues to be one of the most active areas within the field of psychology. It is important not only for childhood but also for adult relationships, as psychoanalysts have suggested in their work on object relations. Discussions between attachment and temperament theorists have often involved controversy. Some attachment theorists have argued that quality of attachment is a relational construct, operating at the level of the dyad and independent of temperament. Taking an equally extreme position, some temperament theorists have argued that attachment measures are simply alternative assessments of infant temperament. In this section we consider ways in which attachment and temperament constructs may be integrated through developmental study.

Satinoff (1982, p. 244) wrote, "An organism's behavior at a particular time depends on the state of its nervous system, the stimuli in its immediate environment, its past individual history, and the evolutionary history of its species." An important aspect of this statement is the conjunctive relationship among its clauses, and a modified version of her analysis can be applied to the development of attachment. Bowlby (1969) noted the adaptive function of proximity maintenance in the protection of human young, and simply identifying humans as a social species suggests the evolutionary function of systems serving to keep individuals physically and emotionally close to others. Bowlby (1969) also identified aspects of a child's state (e.g., pain, discomfort, or distress) and situational novelty that influence attachment-related behavior. If we add the contributions of individual differences in temperament to this analysis, we note that some infants are more distress-prone than others, as indicated by their threshold and intensity of responsiveness across situations (Rothbart, 1989b). If this distress-proneness makes a contribution to the infant's state, then the nature of children's experiences in situations relevant to attachment will differ (van den Boom, 1989). Temperament can thereby be seen as one of several contributions to the individual's state in the development of attachment.

If differences in distress-proneness contribute to the development of attachment, measures of distress-proneness in the newborn period ought to predict later measures of attachment security, at least to a degree. This has proven to be the case, although with some interesting differences in outcome depending on the culture in which the infant was raised. Distress-prone infants from northern Europe (Germany and the Netherlands) were more likely than less distress-prone infants to be later identified in Category A, that is, as avoidant or unlikely to actively use the mother for comfort during reunion after her absence during the Strange Situation, perhaps even to ignore her upon her return (Grossman, Grossman, Spangler, Suess, & Unzner, 1985; van den Boom, 1989). Distress-prone infants from Japan and to a lesser degree the United States, on the other hand, were more likely to be identified in the insecure–ambivalent Category C (Goldsmith & Alansky, 1987; Miyake, Chen, & Campos, 1985). Infants in Category C tend to express negative affect toward the mother upon her return and do not appear to gain comfort from her during reunion. (Infants in secure Category B appear to gain comfort from the mother when she returns, even though they may at first demonstrate distress upon her return.) These cross-cultural findings are very interesting because they suggest that although temperament makes a contribution toward behavior in the Strange Situation, experiences of infants related to their cultural group are also important. Van den Boom (1989) investigated this idea in more detail in her longitudinal studies of children in the Netherlands.

Van den Boom (1989) carried out two longitudinal studies involving lower socioeconomic status infants and their mothers. In the first, she studied the degree to which attachment classification at 1 year might be predicted from (a) infants' distress-proneness at 15 days (assessed during behavioral observation) and (b) mothers' sensitivity and responsiveness to the child (assessed in multiple home observations during the first year). She found that temperamental distress-proneness predicted later attachment classification, especially the avoidant category, but was not able to find that mothers' behavior predicted their later classification. Although van den Boom could have stopped at this point and concluded that a temperament interpretation had prevailed in the conflict between temperament and attachment theories in predicting later child behavior, she looked at her observational data more closely. In these data, she found that for more distress-prone infants, mothers tended over time to increasingly ignore their children and to play little with them.

She therefore designed another study in which distress-prone infants and their mothers were assigned to quasi-experimental conditions. Beginning when the infants were 6 months old, mothers in the experimental group were given a training program in which they learned how to soothe their infants and how to play with them. In comparison with control groups, infants in the experimental group came to engage in more positive affect and less negative affect in the home and showed greater involvement with their mothers. By 12 months they were also much less likely to be categorized as insecurely attached in the Strange Situation (68% secure classification for the experimental infants; 28% for control infants). The experimental infants also exhibited more sophisticated levels of exploratory play with objects, even when their mothers were not present.

Taken separately, van den Boom's first study indicates the strength of temperament in predicting later attachment security

under normally occurring conditions. The second study stresses the influence of maternal skills when training is added. Taken together, however, they illustrate the way in which an interaction between infant predisposition and mother behavior may develop into a trajectory of experience for the child, with important outcomes.

Van den Boom's (1989) study also demonstrates another relationship between temperament and experience: Qualities of the child appear to elicit reactions from others that influence the child's future behavior. We might think about her finding in this way. Generally, infant crying is perceived as distressful to parents. In response to a videotape of a crying infant, Frodi, Lamb, Leavitt, and Donovan (1978) found that parents of newborns reported feelings of distress and showed both increased diastolic blood pressure and skin conductance. This aversive state in parents is thought to motivate behavior to regulate infant distress via soothing. Individual differences in distress-proneness, however, may have some consequences for this general pattern of parental adaptation. Boukydis and Burgess (1982) demonstrated that the audiotaped distress cry of temperamentally distress-prone infants leads to greater distress-related physiological response in adults as well as to more negative characterizations of the infant. The effect of the temperamentally distress-prone infant on the caretaker may thus serve to push the caregiver away. Caregiver avoidance may then result in an interactive cycle in which infants must develop their own means of regulating distress that do not require the caregiver, leading to the avoidant classification. In Germany, infants also appear to be encouraged to deal with emotional challenges on their own, and this too would likely contribute to the child's attachment classification (Grossman et al., 1985).

Other researchers have noted the ways in which reactions of others to the characteristics of the individual child may lead to a set of experiences that may put the child at risk for psychopathology. Rutter and Quinton (1984) found that children with high negative mood, low regularity and malleability, and fastidiousness were likely to elicit irritability, hostility, and criticism from their parents. Rutter (1990) suggested that this "reflects a pattern in which the children's attributes make them a focus for the discord (thus increasing exposure to the risk variable) and increase the probability that exposure will set in motion a train of adverse interactions that will prolong the risk" (p. 191).

Lee and Bates (1985) observed 2-year-olds identified as difficult (defined to a large degree as distress prone) in home situations in which the mother was trying to control the child's behavior. The more difficult children approached trouble more often and were more likely to resist their mothers' attempts at control. In addition, their mothers both exercised more coercive discipline and gave in more frequently to their children. These interaction characteristics, Bates pointed out, are ones that resemble the coercive cycle of behavior described in aggressive boys and their mothers (Patterson, 1977, 1980).

Eisenberg and Fabes (1992) and Rubin, LeMare, and Lollis (1990) have also proposed models for the development of behavior problems in children that specify early temperamental predispositions. In Rubin et al.'s (1990) model, for example, fussy, overactive, and relatively unsoothable infants are seen to develop into children more likely to be actively rejected by their peers because of their expressions of aggression and hostility against other children. The outcome of these experiences may include externalizing problems and possibly dropping out of school.

In our brief discussion of temperament and attachment, we have considered several of Cairns's developmental principles. Most obvious are the bidirectional effects of the infant–caregiver relationship. The infant's temperamental reactivity may serve to push the caregiver away, and with the caregiver less available, the child may develop soothing strategies that do not involve the caregiver. Mothers of distress-prone infants can nevertheless learn to effectively regulate the infant's distress and pleasure to foster involvement of the child with the parent. Multiple levels of analysis are also involved. Temperamental irritability in the newborn refers to a process within the child. However, the development of attachment is a dynamic social process that, depending on the nature of the infant–caregiver relationship, can result in qualitatively quite different outcomes. Finally, the infants' responsiveness to maternal absence and reunion are, above all, adaptive, with responses reflecting learned contingencies between infant and caregiver behavior.

Development of Self-Regulation

As infants develop, they become increasingly able to regulate their own internal states and reactions to external signals. Important in the infant's developing self-regulatory abilities is the maturation of attentional mechanisms (Rothbart & Posner, 1985). An early-developing attentional orienting network appears to be important for directing attention to relevant locations, binding information to spatial locations so as to produce object perception, and selecting a relevant scale for examining visual input (Posner & Petersen, 1990; Rothbart, Derryberry, & Posner, in press). Developmentally, this attentional network shows considerable maturation around the third month of life.

After one month of age and prior to the maturation of the orienting network, infants appear to have difficulty disengaging from a visual stimulus, even when the stimulus becomes distressful. Johnson, Posner, and Rothbart (1991) found that by 4 months, infants were more likely to be able to disengage from a central stimulus to orient to a peripheral target. Moreover, the ability to disengage was negatively correlated with parent reports of infant fear and frustration and positively correlated with parent reports of soothability in the Infant Behavior Questionnaire (Rothbart, 1981). These infants may thus be able to use attentional distraction to soothe themselves.

Caregivers also use distraction methods to soothe their infants. Harman, Rothbart, and Posner (1992) found that after 3- and 4-month-old babies had become distressed, presenting interesting objects caused them to orient to the object and to cease vocal displays of distress. Once the object was removed, however, the babies returned to their initial level of distress. This cycle continued over numerous presentations and removals of interesting objects. Thus, early developments in attentional orienting have implications for the modulation of the expression of distress.

By 4 to 6 months of age, relatively stable individual differences in a set of temperament characteristics related to the broad dimension of surgency or extraversion emerge. At 2 to 3 months, infants show behavior patterns indicative of positive

emotionality, including smiling and vocalization in social contexts. Although some researchers have labeled this behavior pattern *sociability* (e.g., Buss & Plomin, 1975, 1984), smiling and laughter also occur in response to novel and exciting objects (Bradley, 1985). Closely related to positive emotionality are individual differences in approach tendencies and activity level. Approach tendencies are evident, for example, in infant latency to reach for and grasp objects (Rothbart, 1988). Two important implications are apparent with respect to temperamental approach. First, infants with strong approach tendencies are engaging in more active exploration of their environment. They will therefore likely become more familiar with a wider range of people, objects, and events and tend to experience many of these as pleasurable and safe. In addition, individual differences in positive emotionality are related to reward sensitivity, which may mediate learning to conditioned signals of reward (e.g., Gray, 1982).

During the second half of the first year, another important self-regulatory capacity appears to emerge. By 7–8 months, infants begin to display behavioral inhibition in response to novel objects. Infants who were previously relatively quick to approach and grasp unfamiliar objects are now wary of novelty and display caution in their approach. This reaction to novelty may serve as a prototype for anxiety (Rothbart, 1989b; Suomi, 1986). In Gray's (1987) model of anxiety, individual differences in fear proneness or anxiety result from differential sensitivity of an underlying behavior inhibition system. Activation of this system, in response to conditioned signals of punishment, frustrative nonreward, and novelty, results in the experience of fear or anxiety, inhibition of ongoing behavior, increased arousal in preparation for behavioral response, and heightened attention to the environment (Gray, 1987).

Temperamentally fearful children will be more likely to inhibit behavior in response to novel objects presented to them in the laboratory and to be slow and cautious in approaching these objects (Rothbart, 1988). Kagan and his colleagues have referred to this phenomenon as *behavioral inhibition* (Kagan, Reznick, Clarke, Snidman, & Garcia-Coll, 1984). They noted that whereas children, as a function of their social-learning histories, may be shy in the presence of strangers, timid or fearful in the presence of novel objects or situations, or reluctant to try unfamiliar foods or activities, temperamentally inhibited children appear to be biologically prepared to inhibit response across all these stimulus classes (Kagan, Snidman, & Arcus, 1992). Their view is supported by evidence for the heritability of behavioral inhibition in twin studies (Matheny, 1989), heightened sympathetic reactivity to stress in behaviorally inhibited children (Kagan, Reznick, & Snidman, 1988), and differences in baseline levels of cortical activation between temperamentally inhibited and uninhibited 3-year-olds (Davidson, 1992).

Effortful Control and Development

A special class of self-regulatory mechanisms appears to accompany the maturation of attentional mechanisms beginning late in the first year of life (Posner & Rothbart, in press; Rothbart, 1989b). Luria (1973) distinguished between the orienting of attention, which he thought of as involuntary and innate, and a higher form of volitional attentional control developing later, which he believed had social roots. Rapid development at the end of the first year of networks involving frontal brain systems and the prefrontal anterior cingulate gyrus appear to be important in coordinating and controlling attention (Posner & Rothbart, 1992). Prior to 8 or 9 months of age, infants have great difficulty inhibiting their grasping for an object placed behind a plexiglas barrier. Late in the first year, however, infants are able to begin to coordinate attention and movement to reach around the barrier to grasp the object (Diamond, in press). In a similar situation for adults, interference in conflict trials in the Stroop color naming task are marked by increased activation of the anterior attention networks as indicated through positron emission tomography (Posner & Rothbart, 1992). As we shall see, the child's ability to effortfully and voluntarily coordinate attention to think and perform in a means–ends fashion performs important functions in the child's social development.

With the development of attentional self-regulation, the child is increasingly able to inhibit prepotent responses, such as striking out at other children or grabbing others' possessions. By the age of 2 years, children begin to develop normative standards for behavior, that is, to understand what is and is not appropriate (Kagan, 1989). In addition, they begin to understand that their actions have effects on the psychological experiences of others. As described by Zahn-Waxler and colleagues (Zahn-Waxler, Cole, & Barrett, 1991; Zahn-Waxler, Radke-Yarrow, & King, 1979), caregivers often promote the development of empathy and conscience by dramatizing to their children how their transgressions have affected the victim. Processes such as these speed the development of normative standards. However, children differ in their abilities to effortfully inhibit prepotent responses. In a questionnaire study of parent-reported temperament and social behavior, we found individual differences in effortful control to be correlated with characteristic levels of both guilt/shame and empathy (Rothbart et al., 1994).

Kochanska (1991) offered a conceptual and empirical account of the development of guilt and conscience that involves the two temperamental control systems we have described in this article—fear and effortful control (see also Ahadi & Rothbart, in press; Rothbart, 1989b). These temperamental processes are seen to operate in interaction with parental socialization practices to promote moral behavior in the developing child. The first temperamental mechanism involves negative affect and promotes internalization of conscience by associating distress, guilt, and anxiety with wrongdoing. The second temperamental process involves a superordinate self-regulatory ability that enables the child to effortfully or willfully inhibit a forbidden impulse, refrain from wrongdoing, and to respond instead in an acceptable or desired manner (Kochanska, 1991).

Dienstbier (1984) had previously suggested that the internalization of moral behavior is strongly influenced by the first temperamental control mechanism, the individual's susceptibility to conditioned fear or anxiety. He also suggested that lower thresholds for experiencing the internal cues of discomfort associated with fear or anxiety would be associated with a greater likelihood of perceiving the cause of those feelings as being internal, thereby promoting the internalization of moral standards. Kochanska (1991) tested this hypothesis by taking behavioral measures of anxiety or inhibition, children's compliance with their mothers' demands, and parental discipline for a

sample of 2-year-old children. When the children reached 8 to 10 years of age, they were administered a measure of internalized conscience that assessed their affective and moral reactions to stories with moral themes.

Kochanska (1991) found that children of mothers who predominantly used verbal prohibitions (rather than a physical discipline style) when they were 2 were more likely later to react to moral stories with an affective or moral orientation, themes of reparation, and intense discomfort. This pattern of relationships held, however, only for children who at age 2 had been behaviorally inhibited and anxious. Kochanska also found that the children's self-regulation, as indicated by their compliance with maternal demands at age 2, predicted their performance on the measures of conscience at age 8 to 10. We have also found that concurrent temperament measures of negative affectivity and effortful control account for unique variation in parent ratings of both guilt/shame and empathy in a sample of 6- and 7-year-olds (Rothbart et al., 1994). Recent evidence also suggests that the affective developmental pathway to conscience may be more pronounced for girls than boys (Kochanska et al., 1993).

These findings illustrate several developmental processes. Whereas we saw in the attachment discussion that extreme levels of negative affect may predispose the child to develop an avoidant behavioral style, here we see that the absence of anxiety may impair internalization of moral standards for behavior. Child temperament and parental socialization practices are again shown to interact developmentally. For example, parents may find that less coercive discipline techniques, such as appeals to the child's ability to experience distress in response to antisocial behavior, are required to regulate moral behavior in the more fearful or inhibited child. However, such appeals will not be effective if they are not used by the parent, and the appeals may be less effective for less inhibited children, perhaps encouraging more coercive discipline techniques by parents. These techniques in turn may develop into a more confrontational interactive pattern and diminished internalization of moral standards.

That conscience can also be fundamentally influenced through experience is illustrated in the work of Dodge, Bates, and Pettit (1990) on the development of violent behavior. They found that children who had been physically abused early in childhood were significantly more aggressive than their nonharmed peers. This relationship held even when family ecological variables and child biological variables (including retrospective reports of temperament) were covaried. Early experience can therefore be seen to have important consequences for the functioning of the child, even when the child is not temperamentally at risk for such behavior patterns (although the latter point was not given a strong test with prospective measures). Dodge et al. also found, in addition to elevated levels of teacher-rated, peer-rated, and directly observed aggression, that children who had been physically harmed early in childhood evidenced biases in the processing of social information. Previously harmed children were less able to attend to appropriate social cues, were more likely than unharmed children to make hostile attributions of intent from others, and generated fewer competent solutions to interpersonal problems (Dodge et al., 1990). Thus, abused children may develop a mental representation of the world that leads them to construe hostile intent in others, justifying aggression in response to that perceived intent.

Temperamental processes may predispose children to develop such personality patterns as mastery motivation and achievement. For example, children high in effortful control might be better able to use attention to generate possible solutions, inhibit task-irrelevant thoughts, and to persist in response to failure feedback. Shoda, Mischel, and Peake (1990) did a 10-year follow-up study of adolescents who had participated as preschoolers (mean age 4 years and 4 months) in delay-of-gratification experiments. They found that children who were able to self-impose delay of gratification (through attentional strategies) when rewards were exposed during the waiting period were rated by their parents 10 years later as characteristically more persistent, planful, attentive, and able to concentrate and as using and responding to reason. Delay times in this condition also correlated with later performance on the Scholastic Aptitude Test.

Evidence to suggest that these processes may be temperament-based can be found in the behavior of infants. Messer et al. (1986) found that infant persistence and investigation in play tasks observed at 6 and 12 months predicted their intellectual ability at 30 months. This relationship was observed even when the level of intellectual functioning during infancy was covaried. Infant mastery behavior was also related to childhood performance in developmentally meaningful ways. General exploratory behavior at 6 months predicted later ability, and task-directed means–ends behaviors at 12 months also predicted later ability.

Both positive and negative affectivity can also be expected to contribute to achievement-related motivational patterns. Connections between positive affect and potential rewards associated with performance would be fairly straightforward, and negative affect could also affect mastery motivation through frustration and discouragement or anxiety. Gray's (1982, 1987) theory of anxiety suggests that, when individuals perceive cues of punishment, nonreward, or novel stimuli in their environment, they may cease ongoing behavior and heighten monitoring of the environment. They may then have difficulty responding to complex stimuli because much of their available resources are allocated toward monitoring threats. Effects of anxiety may not necessarily be limited to external signals. As noted by Wine (1971, p. 92), "the highly test-anxious subject is internally focused on self-evaluative, self-deprecatory thinking, and perception of his autonomic responses." Cognitive resources may thereby be relatively unavailable for effective problem-solving strategies, creating an experience that may feel like lack of ability to the child.

The development of self-regulation encompasses the entire range of developmental principles. First, the development of self-regulation involves an organized system of psychological and neurophysiological processes, developing over time as a function of maturation and experience. For example, the development of the anterior attention system allows the young child to effortfully use attention to inhibit reactivity in compliance with parental demands, eventually internalizing standards for moral behavior in the development of conscience. In addition, the development of self-regulation is highly social in nature. Children learn moral behavior from parental reactions to mis-

behavior and parental modeling of correct or desirable behaviors. Parents also dramatize the effects of transgressions to facilitate internalization of moral behavior in their children. Peers may reject children who are unable to inhibit inappropriate behaviors. Analyses at multiple levels aid in our understanding of the development of self-regulation. From the neurophysiology of attentional structure and function to the social interactions of children with caregivers and peers, we are beginning to understand how attentional mechanisms mediate the effective learning of moral behavior as conveyed through social interaction.

Environmental Fit

In Cattell's theory, an individual's performance in a given situation will be a function of his or her standing on a constellation of traits, abilities, and motivational characteristics and of the degree to which the corresponding characteristics of the situation "press" upon the individual's characteristics (Cattell, 1957). In the simplest form of his behavior specification equation, if a given situation does not press for a trait, the individual's standing on that trait will be irrelevant in the production of performance. As psychological press on a given trait increases, the individual's standing on that trait will become increasingly important in producing individual performance, with individuals high on the trait sensitive to increasing environmental demands and those low on the trait relatively impervious to such demands.

For example, although a young child may be behaviorally inhibited, parents may work to minimize separation, to regulate exposure to novel objects, people, and situations, and to adapt situations to their child's distress and objections to such exposures. To put it another way, parents may structure the environment to regulate their child's temperamental reactivity. The child does not then characteristically display behavioral inhibition; in this context, fear is irrelevant to the production of thought and behavior. However, when the child reaches school age, she or he will be confronted with an environment characterized by parental separation and novelty of people, situations, and tasks. The environment will also be relatively fixed and not likely to bend to the child's temperamental reactivity.

Even in its simplest form, Cattell's model allows for a wide dynamic range of person–environment interaction. A similar concept in temperament theory has been referred to as *goodness of fit* (Lerner & Lerner, 1983; Thomas & Chess, 1977). In the model of Chess and Thomas (1986), goodness of fit describes situations in which there is a match between the temperamental characteristics of the child and the expectations of others or the demands of the situation or both. A good fit would be predictive of a more favorable mental health outcome; a poor fit, or incompatibility, would be predictive of impaired function and the risk of development of behavior disorders (see also Bem & Funder, 1978, for a discussion of template matching).

Within American society generally, a positive and outgoing temperament may provide a match between the child and many situational requirements. However, some aspects of temperament conducive to the development of social competence may not be conducive to the development of educational competence. As hypothesized by Martin (1989), the classroom "environment rewards the child who is able to remain as physically immobile as is possible, since gross motor movement may impede the attention to learning materials of the child who is moving, and/or it may create environmental stimulation (e.g., noise) that disturbs the attention of other students" (p. 451). Martin reported (a) negative longitudinal relations between temperamental activity level and distractibility and subsequent reading and math achievement and (b) positive relations between temperamental persistence and subsequent achievement. There may also be a relationship between child temperament and teacher perceptions of the child's capacity to learn, although this relationship has yet to be firmly established (Keogh, 1989; Martin, 1989).

Temperament models that explicitly tie the action of traits to environmental circumstances are in accordance with the principle that individuals are continuously active and adaptive throughout the lifespan, with adaptation occurring relative to particular temperamental characteristics and to the conditions requiring adaptation. These models further acknowledge that psychobiological systems are not just organismic but also extend to include the individual's physical and social world. In the following section, we illustrate how consideration of a systems approach may result in the identification of different developmental pathways toward risk of psychopathology.

Developmental Pathways

Temperament may predispose individuals for risk or serve as protective factors for the development of psychopathology in a number of ways. Elsewhere, we have noted numerous ways in which temperament may lead to risk or protective conditions (Rothbart, Posner, & Hershey, in press). Simplest among these, extremes in temperament may be related to risk for development of psychopathology, for example, in relationships between behavioral inhibition and the development of anxiety disorder (Hirshfeld et al., 1992). Very high levels of surgency may also be related to hyperactivity, very low levels to anhedonia. Extreme forms of temperamental functioning may not bear much resemblance to functioning in the normal domain and therefore may be thought of as qualitatively different, resulting in categorization as a type of disorder.

Psychopathology may also represent extremes in a constellation of temperamental processes. For example, in attempting to relate psychopathy to the Big Five dimensions of normal personality description, Harpur, Hart, and Hare (1993) found strong Agreeableness × Neuroticism and Agreeableness × Conscientiousness interactions. This led them to suggest that extreme hostility, indicated by very low agreeableness, tends not to result in psychopathy unless it is accompanied by absence of anxiety and absence of conscientiousness (an adult analog of effortful control). In other words, extreme hostility in the absence of inhibiting controls either through fear or effortful inhibition of prepotent aggressive responses may result in the antisocial and undercontrolled behavior characteristic of psychopathy.

Depressive Mood

Behavior problems develop disproportionately in boys, with antisocial behavior generally found at higher rates for males

than for females (Eagly & Steffen, 1986) and girls found to be at greater risk than boys for depression (Zahn-Waxler et al., 1991). Research results currently describe an interplay between temperament, social experience, and gender in the development of depressive mood, with two recent studies suggesting that antecedents for depressive mood may be somewhat gender specific (Block, Gjerde, & Block, 1991; Patterson & Capaldi, 1990).

Block et al. (1991) described differential developmental pathways to depression as a function of gender. For boys, depressive mood is predicted by a pattern of poor socialization, aggression, and self-aggrandizement, coupled with relatively low intelligence. Block et al. suggested that undercontrolled aggressive responses to others may lead boys to peer rejection, loneliness, and depressed mood. Relatively lower intelligence may also result in greater negative feedback from the environment, decreasing the child's sense of self-worth. Consistent with Block et al.'s (1991) findings, Patterson and Capaldi (1990), in research with two samples of fourth-grade boys, found support for a model in which peer rejection related to undercontrolled behavior was a central mediating influence on depressed mood. The development of depressive mood in girls, on the other hand, appears to be related to excessive control. These girls tend to be oversocialized and to internalize negative affect, and they also tend to be more intelligent. Block et al. (1991) suggested that as bright, hard-working girls become increasingly aware of the criteria by which society evaluates them and limits the avenues available to them, they may begin to feel their situation to be increasingly hopeless. Zahn-Waxler et al. (1991) also suggested a developmental model that highlights vulnerability to depression in highly socialized girls.

Drug Use

Block, Block, and Keyes (1988) observed relationships between personality characteristics of children assessed at 3 and 4 years of age and their drug use at 14 years of age. Children were rated on the California Child Q-sort by nursery school teachers at ages 3 and 4. A composite measure of child personality was then developed by aggregating these Q-sorts, which were then related to Q-sorts of prototypically ego-resilient and ego-undercontrolled children as determined by professional judges. An ego-resilient child was described as resourceful, curious and open to new experience, lively, self-reliant, and able to recover after stressful experiences (Block et al., 1988). An ego-undercontrolled child was described as unable to delay gratification, possessed of rapid personal tempo, emotionally expressive, emotionally labile, and easily irritated or angered.

Determination of drug use was made at age 14 in a clinician's interview. Because the observed relationships for marijuana use and hard drug use were similar, we review here only findings for marijuana use. Girls who later used marijuana were characterized in nursery school as low in ego resiliency and ego control. Adolescent boys who used marijuana were described in nursery school as being chiefly ego undercontrolled. For boys, early-childhood ego resilience was unrelated to later marijuana use.

There also appeared to be differential effects of environment on girls and boys. Although no systematic environmental effects as assessed by the Child Rearing Practices Report were related to later adolescent drug use for boys, there were substantial effects for girls. Mothers of girls who were later likely to use drugs conveyed highly negative feelings toward their daughters. For example, they reported having a sense of disappointment in the daughter, few expectations for her, frequent feelings of anger toward her, and a feeling of having given up personal interests on her behalf. Block et al. (1988) also characterized the family environments of these young girls as permissive, relaxed, and unconventional, where traditional values were not emphasized. If, as we have suggested, effortful control facilitates the development of conscience, these young girls may have been temperamentally less prepared to internalize moral standards, but this characteristic would have been exacerbated by an environment that tended not to encourage internalization of these standards.

The role of gender in the development of personality leads us to consider several of Cairns's developmental principles. First, whereas depressive symptoms in the young men and women in Block et al.'s (1991) study were assessed with the same instrument, the temperamental and environmental factors leading to adolescent expression of depression were different in important ways (Principle 5). The relation between childhood factors and adolescent and adult functioning suggests a fundamental continuity in development, with social and emotional development at earlier ages laying the groundwork for later emotional functioning (Principle 3). It was not, however, early childhood depressive mood that predicted later symptoms but rather undercontrolled and inappropriate social behavior in boys and oversocialized behavior and intelligence in young girls. Consideration of Principles 3 and 5 highlights the holistic functioning of organized systems in personality development (Principle 1). A narrow consideration of isolated processes is not likely to provide us with a rich understanding of social and emotional development; instead, a multivariate analysis of individual and environmental factors in dynamic interaction is required (Ahadi & Diener, 1989; Cattell, 1957).

Alcohol Abuse

As previously noted, different constellations of temperamental processes may result in pathologies that appear similar in their outward manifestations. Cloninger (1987) described two types of alcoholism based on differential constellations of trait scores in his three-dimensional model of personality. Type 1 alcoholics are characterized by an anxious personality profile, with low scores on Cloninger's dimension of novelty seeking and high scores on his dimensions of harm avoidance and reward dependence. Type 2 alcoholism, on the other hand, is characterized by an antisocial personality profile, with high scores on novelty seeking and low scores on harm avoidance and reward dependence. Although superficially these two types of individuals suffer from the same disorder, the nature of the psychopathology is quite different. Type 1 or anxious alcoholics tend not to engage in spontaneous alcohol seeking or to get in fights or be arrested while drinking. They do tend to develop psychological dependence on alcohol, however, and to experience guilt and fear about their alcohol dependence. This pattern is reversed in Type 2 alcoholics.

Cloninger (1987) noted that the neurophysiological and neurochemical processes underlying these subgroups are different, that the motivational patterns underlying alcohol consumption

are different, and that it is likely that treatment interventions will also need to be different. In addition, the two subgroups follow different developmental pathways (Bohman, Sigvardsson, & Cloninger, 1981; Cloninger, 1987; Cloninger, Bohman, & Sigvardsson, 1981). In a large-scale adoption study, subjects were considered genetically at risk for Type 1 (anxious) alcoholism if a biological parent had an adult onset of alcoholism without extensive criminality. They were considered genetically at risk for Type 2 (antisocial) alcoholism if their biological father received treatment for alcoholism with accompanying extensive criminality beginning in adolescence or early adulthood (very few biological mothers met criteria for Type 2). Subjects were considered environmentally at risk for alcoholism if their adoptive father was employed in an occupation in which there would likely be exposure to patterns of heavy recreational drinking.

Genetic factors of antisocially related alcoholism increased the risk for later development of Type 2 (antisocial) alcoholism in males, regardless of their environmental background, but neither genetic background nor environmental background alone was associated with increased risk for development of Type 1 (anxious) alcoholism. Males with a Type 1 genetic risk who also grew up in a home in which heavy drinking was likely were, however, at significantly greater risk for development of later Type 1 alcoholism. In females, only the Type 1 genetic risk predicted Type 1 alcoholism. Again, what may appear to be the same problems may be similar in only superficial properties. Whereas alcohol abuse may be a common presenting symptom, the adaptive mechanisms underlying such abuse may be quite different. In considering developmental pathways to risk of alcoholism, multiple levels of analysis are involved, from genetic predispositions to characteristics of the home environment.

Future Directions

Although the research and theory we have reviewed here was in some cases generated decades ago, we nevertheless view this article as pointing to future directions for research on personality. Margaret Mead (1972) stated this position plainly over 20 years ago:

> Gone are the days, I hope, when students would rebel when I talked in one lecture about innate capacities and individual differences and in the next about the different way in which character is formed in different cultures systematically and how different the results were. Someone was sure to go away muttering: "She can't have it both ways." But, of course, we can. (Mead, 1972, p. vii)

Temperament and trait models of personality and personality development are not static, as was implied by Mischel's (1968) review. Rather they are dynamic, interactive, and conform to developmental principles as outlined by Cairns (1979). Static representations of gene and brain function may have been responsible for creating a static representation of temperament. However, such representations are proving to be false, even at molecular levels, with genetic and neural processes proving to be as flexible and dynamic as the individuals within whom they reside (Black & Greenough, 1991; Gottesman & Goldsmith, in press). We believe future studies of personality development will increasingly take temperament into account, will be respectful of the processes and dynamics of personality development, and will reflect the models of adaptive, organized systems at the biological, personal, and social levels of analysis. We believe this future will be an exciting one.

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